Contest-Aware, Event-Grounded Retrieval-Augmented Generation for Regional Situation Awareness

Brief topic description for Program Officer discussion · FY27 ONR Young Investigator Program (N0001426SF004)

PI: Xinya Du, Assistant Professor of Computer Science, The University of Texas at Dallas · xinya.du@utdallas.edu · xinyadu.github.io
Program Officer: Dr. Rebecca Goolsby, ONR Code 341, Analytics, Modeling and Simulation (Local and Regional)
Proposed S&T Focus Area: Human and Biological Systems — Warfighter Decision Superiority and Readiness; secondary: Artificial Intelligence and Autonomy — Adversarial AI & Autonomy

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Problem. Regional situation awareness increasingly depends on synthesizing news, social media, official statements, and field reports, and retrieval-augmented generation (RAG) is an attractive way to do it. Standard RAG, however, treats retrieved passages as independent evidence: it counts copies as confirmations, loses the timing of corrections, and can be steered by coordinated messaging. After a typhoon, for example, hundreds of posts may claim that a city's water supply is contaminated, most copied from one fabricated account, while an official test showing the water is safe appears in two outlets a day later. A summary built from the most frequently retrieved passages reports contamination as widely confirmed. In a contested information environment, this is the normal condition, not an edge case.

Central hypothesis. Grounding RAG in *contest-aware event graphs*, which record for each claim its sources, copying relations, corrections, and status (supported, contested, or unverified), will yield situation assessments with fewer unsupported claims, faster uptake of corrections, and lower susceptibility to coordinated deception than passage-level RAG, without losing coverage of routine events.

Research thrusts.

- **Contest-aware event graphs.** Extract who did what, where, and when from heterogeneous streams; link reports that describe the same event; and detect copying, coordinated amplification, corrections, and retractions from text reuse, timing, and shared origins. Keep credible competing versions instead of merging them into a false consensus.
- **Generation that preserves contestation.** Retrieve through the event graph and generate assessments in which each claim carries its status and its independent sources. Adapt the fine-grained feedback methods we developed for vision-language models (FaithScore, FGAIF) to text RAG, training the generator to mark status correctly and to request an independent source when a consequential claim is contested.
- **Ground-truthed synthetic information environments.** Generate report streams from event schemas with known histories, and plant deception (amplification, timing manipulation, fabricated corrections) under varied adversary strategies. The same environments support evaluation and what-if analysis for exercises and wargames.

What is new. Most RAG pipelines treat retrieved sources as independent, event extraction rarely records whether a claim is contested, and much misinformation detection classifies posts or threads without event-level status. We track contestation at the event level and carry it through retrieval and generation.

Evaluation. GDELT event streams, public humanitarian situation reports with known timelines, and the synthetic environments above. Measures: claim-level support, accuracy of claim status, time to incorporate a correction, rate of repeating planted false claims, and coverage of routine events. Baselines: standard RAG, iterative RAG with LLM self-critique, and source-count aggregation.

Naval relevance. Navy and Marine Corps forces conducting humanitarian assistance or littoral operations must act on a regional picture assembled from open sources that adversaries can manipulate. Planners need to know which claims about roads, ports, water, and fuel are supported, contested, or unverified, and by how many independent sources. The topic addresses the program's interests in AI-enhanced retrieval-augmented generation for warfighter performance, modeling and simulation for operations in the information environment, and problems of deception and obfuscation.

PI foundation. Event extraction by answering questions (EMNLP 2020); template filling with generative transformers (NAACL 2021); MEQA, multi-hop event-centric question answering (NeurIPS 2024); FaithScore and FGAIF, fine-grained hallucination evaluation and AI feedback for vision-language models (EMNLP Findings 2024; TMLR 2025). The PI's NSF CAREER award (2024) extracts consistent event graphs from documents; contest-awareness and adversarial evaluation are new here. As a postdoc, the PI led system integration for the RESIN team in DARPA KAIROS (RESIN-11, NAACL 2022 Demo, first author).